



CORE PROJECT DOCUMENTATION PACK

Siyani — AI Assistant Strategy

Document 03 of 08 · PRODUCT & AI

M A D HALO Technologies



What Siyani Is

Siyani is the AI assistant at the heart of HI NGE SIYANI — the interface people talk to when they want help with education, opportunities, or government/public-service information. Siyani exists to make that help available in the language a person is actually comfortable using.

What Siyani Should Be Able To Do

- Understand a user's question, in their chosen language, and respond helpfully and accurately in that same language
- Guide users through education-related questions: pathways, requirements, applications, opportunities
- Help users understand government and public-service processes and information
- Ask clarifying questions when a request is ambiguous, rather than guessing
- Personalise responses based on what's relevant to the user (e.g. their language, stated goals, or profile information), without overstepping into assumptions the user hasn't confirmed
- Clearly communicate when it doesn't know something or isn't confident, instead of guessing

How Multilingual Understanding Should Work

Multilingual understanding happens on two levels:

1. **Input understanding** — correctly interpreting what the user means, in their language, including local phrasing and terminology (not just a literal, word-for-word reading).
2. **Output generation** — responding in natural, correct language, not a stiff or overly literal translation of an English answer.

Both levels matter. A system that understands a question but replies in awkward, translated-sounding language will still lose the user's trust.

How the User's Preferred Language Works

- Users select a preferred language during onboarding.
- This preference is stored on their profile and used by default in all interactions with Siyani.
- Users can change their preferred language at any time, and the change should apply immediately.

How Users Can Switch Languages Without Losing Context

When a user switches language mid-conversation, Siyani should:

- Continue the same conversation/topic — the underlying context (what the user was asking about) should not be lost



- Respond in the newly selected language from that point forward
- Avoid forcing the user to repeat themselves

This depends on separating "what the user is trying to do" (the context) from "what language it's expressed in" (the presentation layer) — the two should not be tightly coupled in how Siyani is built.

How Siyani Should Handle Uncertainty

Siyani should be built to recognise when it is uncertain — about a fact, a translation, or the right process to describe — and to say so plainly rather than presenting a guess as if it were confirmed. Where possible, it should point the user toward where they can verify the information themselves (an official source, institution, or office), rather than being the sole authority on sensitive or high-stakes information.

This matters most for education and government/public-service topics, where an incorrect answer can have real consequences for someone's application, rights, or opportunities.

How Information Should Be Verified

- Education and public-service information should be sourced from official or otherwise reliable references wherever possible, not generated purely from a general-purpose AI model's background knowledge.
- Information that changes over time (deadlines, requirements, fees, processes) needs a process for staying current — stale information can be as harmful as wrong information.
- Where Siyani is not confident a piece of information is current or accurate, it should say so, rather than stating it as fact.

How Education Information Should Be Handled

Education guidance should focus on helping users understand pathways, requirements, and opportunities clearly — without presenting Siyani as a replacement for official institutional guidance (e.g. a university's own admissions office). Siyani's role is to make the *first step* of understanding easier, and to help users know what to ask or where to go next.

How Government/Public-Service Information Should Be Handled

This is the highest-stakes category, since incorrect information can affect someone's rights, benefits, or legal standing. Government/public-service answers should:

- Be treated with extra caution around certainty and currency
- Point users toward official channels for anything that requires formal confirmation or action
- Avoid presenting Siyani's answer as a substitute for an official decision or document



Future Voice Capabilities

Voice input and output are planned for a later phase. When introduced, the same principles apply: accurate understanding, natural (not robotic or overly literal) responses, and the same care around uncertainty and verification — voice should not be treated as "text with extra steps," since spoken language has its own natural patterns that need to be respected.

What Should NOT Be Claimed About the AI Yet

To keep this document honest and useful over time:

- Siyani should not be described as fully fluent or equally strong across all target languages until that has actually been tested and validated
- Voice capability should not be described as available until it is built and working
- Siyani should not be presented as an official government or academic authority — it is a guide and assistant, not an official record or decision-maker
- No claim should be made that HI NGE SIYANI has trained its own foundation language model — at this stage, it has not

Existing AI Model vs. Multilingual Prompting/RAG vs. Training Our Own Language Technology

These are three different things, and it's worth being precise about which one HI NGE SIYANI is actually doing at each stage:

1. Using an existing AI model

Siyani, at this stage, is built on top of an existing general-purpose AI model (a large language model). This gives strong general reasoning and language ability without needing to build a model from scratch. The trade-off is that the model's knowledge and language quality — especially for under-resourced South African languages — depends entirely on how well the underlying model was trained on those languages, which may be uneven.

2. Multilingual prompting / Retrieval-Augmented Generation (RAG)

This is the realistic near-term strategy for improving accuracy and language quality without training a new model. It means:

- Structuring how Siyani is instructed (prompted) to produce natural, accurate responses in each target language
- Retrieving relevant, curated, verified content (education info, public-service info, glossaries) at the moment of the query, and having the model use that content to answer — rather than relying purely on what the model "remembers"

This approach lets HI NGE SIYANI improve accuracy and language quality steadily, using real curated data, without the cost and complexity of training a new model.



3. Training or fine-tuning our own language technology

This is a long-term, later-phase possibility — training or fine-tuning models specifically for South African languages (text or speech), likely in partnership with universities or language researchers, once there is enough validated data and a clear case for why an existing model plus RAG isn't sufficient. This is not something to attempt immediately, and this document does not recommend it as a near-term step.